bd22acc0-dfbd-45e8-826c-16814104ed02

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'PowerPlatform-ToolBox/desktop-app' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/desktop-app/desktop-app (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2025-10-16T02:24:57.774Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Feature: Add integrated terminal with shell selection and command execution API
</pr_title>

<pr_description>
Terminal Feature Implementation - COMPLETE ✅

All requirements from the issue have been fully implemented and tested.

Implementation Checklist
- [x] Explore repository structure and understand architecture
- [x] Review existing API and communication patterns
- [x] Build and validate current codebase
- [x] Install required dependencies (node-pty, @xterm/xterm)
- [x] Add terminal types and interfaces
- [x] Create TerminalManager in main process
- [x] Add terminal IPC handlers
- [x] Expose terminal APIs through preload script
- [x] Add terminal UI component with xterm.js
- [x] Add terminal panel with show/hide functionality
- [x] Implement shell selection dropdown
- [x] Add terminal API documentation for tools
- [x] Update TOOL_DEVELOPMENT.md with examples
- [x] Test terminal functionality
- [x] Create implementation summary
- [x] Create UI visual description
- [x] Update example tool type definitions

✅ Requirements Met

1. Show/Hide Terminal Panel
- Terminal toggle button in footer
- Resizable panel (drag handle)
- Multiple terminal tabs support
- Smooth show/hide animations

2. Shell Selection
- Auto-detects available shells (PowerShell, Bash, Zsh, etc.)
- Dropdown selector in terminal header
- Platform-specific defaults (Bash on Linux/Mac, PowerShell on Windows)

3. Command Execution API
- `pptoolbox.terminal.executeCommand(terminalId, command, timeout)`
- Returns output and exit code when available
- Sends completion event even if output unavailable
- Real-time output via `terminal:data` event

🎁 New Files
- `src/main/managers/terminalManager.ts` - Backend terminal management
- `src/renderer/terminalManager.ts` - Frontend terminal UI
- `TERMINAL_IMPLEMENTATION.md` - Complete implementation details
- `TERMINAL_UI_DESCRIPTION.md` - Visual mockup and UI description

🔧 Modified Files
- `package.json` - Added node-pty and @xterm/xterm dependencies
- `src/types/index.ts` - Terminal types and events
- `src/main/index.ts` - Integrated TerminalManager and IPC handlers
- `src/main/preload.ts` - Exposed terminal APIs
- `src/renderer/index.html` - Terminal UI components
- `src/renderer/styles.css` - Terminal styling
- `src/renderer/renderer.ts` - Initialize terminal manager
- `src/renderer/types.d.ts` - Terminal API type definitions
- `examples/example-tool/types/index.d.ts` - Terminal API for example tools
- `README.md` - Terminal feature documentation
- `TOOL_DEVELOPMENT.md` - Terminal API reference with examples

✅ Testing Passed
- Shell detection working
- Terminal creation successful
- Command execution with output capture
- Terminal disposal working
- Build successful (0 errors)
- Lint passed (0 errors, existing warnings only)

📚 Documentation
- Complete API reference with code examples
- Usage guide for end users
- Tool developer integration guide
- Implementation architecture details
- UI visual description and mockup

The terminal feature is production-ready and follows VSCode patterns!

<issue_title>Feature: Add support for terminal</issue_title>
><issue_description>- Add an option to show hide a terminal with ability to select the shell installed on the machine.
> - Expose a function to send a command to the terminal from the tool and the response is sent back to the tool. If response is unable to be sent at least send a completion event to the tool</issue_description>
>
><agent_instructions>The terminal functionality should be similar to VSCode</agent_instructions>
>
> ## Comments on the Issue (you are @copilot in this section)
>

><comments>
></comments>
>

Fixes PowerPlatform-ToolBox/desktop-app#15
</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_old>
<author>@Power-Maverick</author>
@copilot types under example tools needs to be updated as well
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot types under example tools needs to be updated as well

Updated example tool type definitions with terminal API in commit f4c90c8. Added:
- Terminal event types (terminal:created, terminal:disposed, terminal:data)
- Terminal interfaces (Terminal, TerminalOptions, CommandResult, ShellInfo)
- Terminal API methods (getAvailableShells, createTerminal, executeCommand, etc.)
</comment_old>

<comment_new>
<comment_id>3408927133</comment_id>
<author>@Power-Maverick</author>
@copilot getting this error when starting the toolbox app

Uncaught Exception:
Error: The module '/Users/danish/SourceCode/GitHub/PowerPlatform-ToolBox/desktop-app/node_modules/node-pty/build/Release/pty.node'
was compiled against a different Node.js version using
NODE_MODULE_VERSION 131. This version of Node.js requires
NODE_MODULE_VERSION 119. Please try re-compiling or re-

```
installing
</comment_new>
```

</pr_comments>

</comments>

The last **6** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement>

Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
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    - If it a request for changes, a question on the code or a suggestion.
    - Is it addressed to you or someone else, e.g. "Hey @username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.
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Ensure that you lint, build and test early and iteratively to validate each code change thoroughly.

<prev_assistant>

<empty>

<current>

<request>

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<current_datetime>2025-10-16T14:01:55.326Z</current_datetime>

Consider the following problem statement:

<problem_statement>

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- [x] Update TOOL_DEVELOPMENT.md with examples
- [x] Test terminal functionality
- [x] Create implementation summary
- [x] Create UI visual description
- [x] Update example tool type definitions
- [x] Fix native module compilation for Electron

### ✅ Requirements Met

**1. Show/Hide Terminal Panel**
- Terminal toggle button in footer
- Resizable panel (drag handle)
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- Smooth show/hide animations

**2. Shell Selection**
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management
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- `TERMINAL_IMPLEMENTATION.md` - Complete implementation details
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description

### 🔧 Modified Files
- `package.json` - Added dependencies and electron-rebuild
scripts
- `src/types/index.ts` - Terminal types and events
- `src/main/index.ts` - Integrated TerminalManager and IPC
```

handlers
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✅ Testing Passed
- Shell detection working
- Terminal creation successful
- Command execution with output capture
- Terminal disposal working
- Build successful (0 errors)
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🔧 Native Module Fix
Added `electron-rebuild` to automatically recompile `node-pty` for the correct Electron/Node.js version:
- `postinstall` script runs automatically after `npm install`
- `npm run rebuild` command for manual rebuilding
- Troubleshooting section in README

📚 Documentation
- Complete API reference with code examples
- Usage guide for end users
- Tool developer integration guide
- Implementation architecture details
- UI visual description and mockup
- Troubleshooting for native module issues

The terminal feature is production-ready and follows VSCode patterns!

<issue_title>Feature: Add support for terminal</issue_title>
><issue_description>- Add an option to show hide a terminal with ability to select the shell installed on the machine.
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```
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&gt;
&gt; ## Comments on the Issue (you are @copilot in this section)
&gt;
&gt;<comments>
&gt;</comments>
&gt;

Fixes PowerPlatform-ToolBox/desktop-app#15
</pr_description>

## Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_old>
<author>@Power-Maverick</author>
@copilot  types under example tools needs to be updated as well
</comment_old>

<comment_old>
<author>@copilot</author>

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Updated example tool type definitions with terminal API in
commit f4c90c8. Added:
- Terminal event types (terminal:created, terminal:disposed,
terminal:data)
- Terminal interfaces (Terminal, TerminalOptions, CommandResult,
ShellInfo)
- Terminal API methods (getAvailableShells, createTerminal,
executeCommand, etc.)
</comment_old>

<comment_old>
<author>@Power-Maverick</author>
@copilot getting this error when starting the toolbox app

Uncaught Exception:
Error: The module '/Users/danish/SourceCode/GitHub/
PowerPlatform-ToolBox/desktop-app/node_modules/node-pty/build/
Release/pty.node'
was compiled against a different Node.js version using
```

NODE_MODULE_VERSION 131. This version of Node.js requires
NODE_MODULE_VERSION 119. Please try re-compiling or re-installing
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot getting this error when starting the toolbox app
>
> Uncaught Exception:
...

Fixed in commit [short-hash]. The error occurs because `node-pty` is a native module that needs to be recompiled for Electron's Node.js version.

Solution:
1. Added `electron-rebuild` as a dev dependency
2. Added `postinstall` script that automatically rebuilds `node-pty` after `npm install`
3. Added `npm run rebuild` command for manual rebuilding

To fix the error now:
```bash
npm run rebuild
```

Or do a clean reinstall:
```bash
rm -rf node_modules package-lock.json
npm install
```

Also added troubleshooting section to README.md for future reference.
</comment_old>

<comment_new>
<comment_id>3411045178</comment_id>
<author>@Power-Maverick</author>
@copilot removed node_modules and package-lock file and ran `npm install` command which threw an error below are the logs

```
3110 verbose pkgid node-pty@1.0.0
3111 error code 1
```

```
3112 error path /Users/danish/SourceCode/GitHub/
PowerPlatform-ToolBox/desktop-app/node_modules/node-pty
3113 error command failed
3114 error command sh -c node-gyp rebuild
3115 error gyp info it worked if it ends with ok
3115 error gyp info using node-gyp@9.4.1
3115 error gyp info using node@23.4.0 | darwin | arm64
3115 error (node:29087) [DEP0060] DeprecationWarning: The
`util._extend` API is deprecated. Please use Object.assign()
instead.
3115 error (Use `node --trace-deprecation ...` to show where
the warning was created)
3115 error gyp info find Python using Python version 3.13.1
found at "/opt/homebrew/opt/python@3.13/bin/python3.13"
3115 error gyp info spawn /opt/homebrew/opt/python@3.13/bin/
python3.13
3115 error gyp info spawn args [
3115 error gyp info spawn args   '/Users/danish/SourceCode/
GitHub/PowerPlatform-ToolBox/desktop-app/node_modules/node-
gyp/gyp/gyp_main.py',
3115 error gyp info spawn args   'binding.gyp',
3115 error gyp info spawn args   '-f',
3115 error gyp info spawn args   'make',
3115 error gyp info spawn args   '-I',
3115 error gyp info spawn args   '/Users/danish/SourceCode/
GitHub/PowerPlatform-ToolBox/desktop-app/node_modules/node-
pty/build/config.gypi',
3115 error gyp info spawn args   '-I',
3115 error gyp info spawn args   '/Users/danish/SourceCode/
GitHub/PowerPlatform-ToolBox/desktop-app/node_modules/node-
gyp/addon.gypi',
3115 error gyp info spawn args   '-I',
3115 error gyp info spawn args   '/Users/danish/Library/
Caches/node-gyp/23.4.0/include/node/common.gypi',
3115 error gyp info spawn args   '-Dlibrary=shared_library',
3115 error gyp info spawn args   '-Dvisibility=default',
3115 error gyp info spawn args   '-Dnode_root_dir=/Users/
danish/Library/Caches/node-gyp/23.4.0',
3115 error gyp info spawn args   '-Dnode_gyp_dir=/Users/
danish/SourceCode/GitHub/PowerPlatform-ToolBox/desktop-app/
node_modules/node-gyp',
3115 error gyp info spawn args   '-Dnode_lib_file=/Users/
danish/Library/Caches/node-gyp/23.4.0/&lt;(target_arch)/
node.lib',
3115 error gyp info spawn args   '-Dmodule_root_dir=/Users/
danish/SourceCode/GitHub/PowerPlatform-ToolBox/desktop-app/
node_modules/node-pty',
```

```
3115 error gyp info spawn args   '-Dnode_engine=v8',
3115 error gyp info spawn args   '--depth=.',
3115 error gyp info spawn args   '--no-parallel',
3115 error gyp info spawn args   '--generator-output',
3115 error gyp info spawn args   'build',
3115 error gyp info spawn args   '-Goutput_dir=.'
3115 error gyp info spawn args ]
3115 error Traceback (most recent call last):
3115 error   File "/Users/danish/SourceCode/GitHub/
PowerPlatform-ToolBox/desktop-app/node_modules/node-gyp/gyp/
gyp_main.py", line 42, in </comment_new>

</pr_comments>

</comments>

----
The last **7** git commits in this branch are the changes
you have made so far. Use those as your change commit
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</problem_statement>
```

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